

Parthiv Paul

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EDUCATION

University of Toronto St. George <i>Bachelor of Computer Science, Specialist Co-op Program CGPA: 3.6/4.0</i>	Toronto, ON <i>Sept 2024 – Apr 2028</i>
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EXPERIENCE

Software Developer, Contract <i>GenLedge</i> – Built a Stripe MCP server (TypeScript, 12 tools) covering charges, invoices, subscriptions, and disputes, enabling AI accounting agents to query live payment data via the Model Context Protocol. – Built a Stripe webhook handler that verified event authenticity and routed 12 event types to specialized AI employee roles, generating real-time general-ledger entries automatically.	Mar 2026 – May 2026 <i>Remote</i>
AI and Automation Extern <i>Outamation</i> – Built NLP and Computer Vision (OCR) pipelines to classify and extract fields from mortgage documents, replacing a manual data-entry step on the program's sample document set. – Improved retrieval relevance by ~25% on a benchmark query set by tuning a Retrieval-Augmented Generation (RAG) system in LlamaIndex, iterating on chunking and custom vector embeddings.	May 2025 – Aug 2025 <i>Remote</i>
Software QA Tester, Intern <i>Chester-Hill Solutions</i> – Wrote automated test scripts covering 50+ user flows across multiple environments, cutting the repetitive manual regression passes the team ran each release cycle. – Investigated and documented defects through API response analysis and root-cause write-ups, giving developers clearer repro steps to turn around fixes faster.	June 2024 – Nov 2024 <i>Remote</i>

PROJECTS

LeaseGuard <i>TypeScript, Next.js, Claude (MCP), Supabase pgvector, RAG</i> – Built an AI agent that analyzes Ontario leases clause by clause, using a 12-tool MCP server (Claude Haiku 4.5) to retrieve statute law and tribunal decisions and return a grounded risk report in under 90 seconds. – Engineered hybrid BM25 and vector retrieval (reciprocal rank fusion) over a 2,372-chunk pgvector legal corpus, with a deterministic TypeScript risk engine for 17 violation types and 206 automated tests.
Palate (UofTHacks 2026) <i>Next.js, TypeScript, MongoDB, WebAuthn, Google Gemini</i> – Built the Next.js frontend and end-to-end passkey authentication (WebAuthn) for a group-dining app whose Gemini engine turns group swipe data into restaurant picks, addressing the 1Password challenge.
Gist <i>TypeScript, React, FastAPI, Python, Google Gemini, Manifest V3, SSE</i> – Built a Manifest V3 Chrome extension with three isolated execution contexts and a Shadow DOM popover, routing all network requests through the background service worker to bypass host-page content security policies. – Streamed real-time AI explanations from a FastAPI backend over server-sent events, with a typed message contract enforced across contexts and full test-first coverage in Vitest and Pytest.
Reflecta <i>Python, Typer, Gemini, Groq, Claude, pytest, AST</i> – Published a Python CLI agent (reflecta on PyPI) that finds coverage gaps, generates pytest tests, and repairs failures, lifting coverage from 64% to 80% on a real service and 40% to 100% on the sample. – Designed a multi-model routing pipeline that uses Gemini for generation, Groq for repair, and Claude for escalation, gated by AST assertion, coverage-delta, and mutation checks across 338 tests.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript (ES6+), Java, SQL, HTML5, CSS
Frameworks: React, Next.js, Node.js, Express.js, FastAPI, TailwindCSS, Zustand, Framer Motion, LlamaIndex, Redux
AI Tools: Claude Code, Claude API, Model Context Protocol (MCP), RAG, Google Gemini, Cursor, OpenAI Codex
Dev Tools: Git, GitHub, Docker, REST APIs, SSE, OAuth 2.0, WebAuthn, CI/CD, Vite, Agile/Scrum

AWARDS

University of Toronto Dean's List (2024-2025, 2025-2026) J.S. McLean Scholarship (UofT)
2nd Place – National Waterloo SAF Financial Literacy Competition (2023)

CERTIFICATIONS

Claude Code in Action (Anthropic) Building with the Claude API (Anthropic)
AI Agents Intensive Course (Google & Kaggle) ChatGPT Prompt Engineering for Developers (DeepLearning.AI)